

Task2

The screenshot displays a development environment with three main components:

- Code Editor (Left):** Shows the `main.go` file with a `func main()` function. It includes a `DELETE /assets/{id}` endpoint and a `POST /assets/batch` endpoint. The code uses `log.Println` for logging and `http.ListenAndServe` for the server.
- Terminal (Bottom):** Shows the output of the `main.go` program. It indicates that the server is listening on `http://localhost:8080` and provides instructions for adding logging middleware and request ID tracking.
- API Documentation (Right):** Shows the API specification for the `DELETE /assets/{id}` endpoint. It includes the request body structure, which is a JSON object with `assets` and `filters` fields.

Bài 2: Batch Create

Yêu cầu: Tạo nhiều assets cũ

API Specification

task1

The screenshot shows a terminal window with the following commands and output:

```
MINGW64:/c/Users/a/Desktop/session2-basic-api

a@DESKTOP-R4B2HM0 MINGW64 ~/Desktop/session2-basic-api
$ curl -X POST http://localhost:8080/assets/batch \
  -H "Content-Type: application/json" \
  -d '{
    "assets": [
      {"name": "test1.com", "type": "domain"},
      {"name": "test2.com", "type": "domain"}
    ]
  }'
{"created":2,"ids":["ce2b7fac-a4ca-4e0a-b066-08b4bd46e1ef","cc0d3418-5eb7-49de-bbdf-83a02c8e9144"]}
```

The output shows that the `POST` request was successful, returning a `200` status code and a JSON response with the `created` count and the `ids` of the created assets.

Task 5

```
a@DESKTOP-R4B2HMO MINGW64 ~/Desktop/session2-basic-api
$ curl.exe http://localhost:8080/health
{"status":"ok","storage":{"type":"in-memory","asset_count":8},"uptime_seconds":1971.4533402,"timestamp":"2026-06-14T15:53:23.8568877+07:00"}

a@DESKTOP-R4B2HMO MINGW64 ~/Desktop/session2-basic-api
$ |
```

Task3

```
MINGW64: c:/Users/a/Desktop/session2-basic-api
a@DESKTOP-R4B2HMO MINGW64 ~/Desktop/session2-basic-api
$ curl.exe http://localhost:8080/health
{"status":"ok","storage":{"type":"in-memory","asset_count":8},"uptime_seconds":1971.4533402,"timestamp":"2026-06-14T15:53:23.8568877+07:00"}

a@DESKTOP-R4B2HMO MINGW64 ~/Desktop/session2-basic-api
$ ID1=$(curl -s -X POST http://localhost:8080/assets -H "Content-Type: application/json" -d '{"name":"test1.com","type":"domain"}' | grep -o '"id":"[^"]*"' | cut -d'"' -f4)
ID2=$(curl -s -X POST http://localhost:8080/assets -H "Content-Type: application/json" -d '{"name":"test2.com","type":"domain"}' | grep -o '"id":"[^"]*"' | cut -d'"' -f4)

a@DESKTOP-R4B2HMO MINGW64 ~/Desktop/session2-basic-api
$ curl -X DELETE "http://localhost:8080/assets/batch?ids=$ID1,$ID2,fake-uuid-123"
{"deleted":2,"not_found":1}

a@DESKTOP-R4B2HMO MINGW64 ~/Desktop/session2-basic-api
$ curl http://localhost:8080/assets/$ID1
{"error":"asset not found"}

a@DESKTOP-R4B2HMO MINGW64 ~/Desktop/session2-basic-api
$ |
```

Task 4:

a@DESKTOP-R4B2HMO MINGW64 ~/Desktop/session2-basic-api

\$ for i in \$(seq 1 20); do

curl -s -X POST http://localhost:8080/assets \

-H "Content-Type: application/json" \

-d '{"name":"concurrent-\$i.com","type":"domain"}' &

done

[1] 983

[2] 984

[3] 985

```
{"id":"ad1d7277-13df-4575-8143-0ceb8582bd56","name":"concurrent-1.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:19.9309376+07:00","updated_at":"2026-06-14T16:00:19.9309376+07:00"}
```

[4] 986

```
{"id":"76ba8de3-8dda-4503-98d8-eb1beffe8f7b","name":"concurrent-2.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:19.9444654+07:00","updated_at":"2026-06-14T16:00:19.9444654+07:00"}[5] 987
```

```
{"id":"f07cebae-45d5-4274-ad16-70355e6b34b3","name":"concurrent-3.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:19.9600866+07:00","updated_at":"2026-06-14T16:00:19.9600866+07:00"}
```

[6] 988

```
{"id":"6e81f368-9fb6-44c2-b939-e6aec47118b2","name":"concurrent-4.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:19.9753323+07:00","updated_at":"2026-06-14T16:00:19.9753323+07:00"}[7] 989
```

```
{"id":"b6d4c5a3-8635-459f-afc3-1ea92a609007","name":"concurrent-5.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:19.9908246+07:00","updated_at":"2026-06-14T16:00:19.9908246+07:00"}
```

```
{"id":"da3fd9a0-8bb9-48db-ba6e-53975ceccb51","name":"concurrent-6.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:20.0053425+07:00","updated_at":"2026-06-14T16:00:20.0053425+07:00"}[8] 990
```

[9] 991

```
{"id":"aa591352-a9e5-4d07-a56a-6d44f2ac0cac","name":"concurrent-7.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:20.0221551+07:00","updated_at":"2026-06-14T16:00:20.0221551+07:00"}
```

```
{"id":"7f1f51a3-3a05-49f2-b13d-c38b2d5900d4","name":"concurrent-8.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:20.0371146+07:00","updated_at":"2026-06-14T16:00:20.0371146+07:00"}[10] 992
```

[11] 993

```
{"id":"1ed9d3ab-69be-453e-aa2f-5ffc6932d44e","name":"concurrent-9.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:20.0540228+07:00","updated_at":"2026-06-14T16:00:20.0540228+07:00"}[12] 994
```

[13] 995

```
{"id":"c7bce939-f1fa-408d-aa71-bade758dd3cb","name":"concurrent-10.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:20.0697264+07:00","updated_at":"2026-06-14T16:00:20.0697264+07:00"}
```

```
{"id":"7b23ed8a-325f-4a94-9079-9e290ffe9661","name":"concurrent-11.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:20.0995354+07:00","updated_at":"2026-06-14T16:00:20.0995354+07:00"}
```

```
{"id":"7d91e239-6426-4d83-9cba-1e1ffc5a0317","name":"concurrent-12.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:20.1295389+07:00","updated_at":"2026-06-14T16:00:20.1295389+07:00"}
```

[14] 996

[15] 997

```
{"id":"b808c45e-3eb9-480a-96b5-79d02ccae12f","name":"concurrent-13.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:20.1459935+07:00","updated_at":"2026-06-14T16:00:20.1459935+07:00"}
```

[16] 998

```
{"id":"6cc6f27f-09fe-46bc-a0cd-cc4254f48f7e","name":"concurrent-14.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:20.1612069+07:00","updated_at":"2026-06-14T16:00:20.1612069+07:00"}[17] 999
```

[18] 1000

```
{"id":"54f97f90-86a7-4fec-b7fc-b35ba9523ad4","name":"concurrent-15.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:20.1769419+07:00","updated_at":"2026-06-14T16:00:20.1769419+07:00"}[19] 1001
```

[20] 1002

```
{"id":"fee2eb14-0bb4-4104-a46f-2e33656d7efd","name":"concurrent-16.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:20.1928529+07:00","updated_at":"2026-06-14T16:00:20.1928529+07:00"}
```

```
{"id":"5e621e52-8c8c-4b65-ac6b-f585bfaf38e7","name":"concurrent-17.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:20.2224471+07:00","updated_at":"2026-06-14T16:00:20.2224471+07:00"}
```

```
{"id":"90e94e24-24b2-4826-b052-d9afacef39ff","name":"concurrent-18.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:20.2374472+07:00","updated_at":"2026-06-14T16:00:20.2374472+07:00"}
```

```
{"id":"f2d42d7e-7d46-4643-9c1d-69105ea1626b","name":"concurrent-19.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:20.2694772+07:00","updated_at":"2026-06-14T16:00:20.2694772+07:00"}
```

```
{"id":"d05a7a5b-b04e-4d5e-965e-ddfda38a7491","name":"concurrent-20.com","type":"domain","status":"active","created_at":"2026-06-14T16:00:20.3003961+07:00","updated_at":"2026-06-14T16:00:20.3003961+07:00"}
```

```
[1] Done      curl -s -X POST http://localhost:8080/assets -H "Content-Type: application/json" -d "{\"name\":\"concurrent-$i.com\",\"type\":\"domain\"}"
```

```
[2] Done      curl -s -X POST http://localhost:8080/assets -H "Content-Type: application/json" -d "{\"name\":\"concurrent-$i.com\",\"type\":\"domain\"}"
```

```
[3] Done      curl -s -X POST http://localhost:8080/assets -H "Content-Type: application/json" -d "{\"name\":\"concurrent-$i.com\",\"type\":\"domain\"}"
```

```
[4] Done      curl -s -X POST http://localhost:8080/assets -H "Content-Type: application/json" -d "{\"name\":\"concurrent-$i.com\",\"type\":\"domain\"}"
```

```
[5] Done      curl -s -X POST http://localhost:8080/assets -H "Content-Type: application/json" -d "{\"name\":\"concurrent-$i.com\",\"type\":\"domain\"}"
```

```
[6] Done      curl -s -X POST http://localhost:8080/assets -H "Content-Type: application/json" -d "{\"name\":\"concurrent-$i.com\",\"type\":\"domain\"}"
```

```
[7] Done      curl -s -X POST http://localhost:8080/assets -H "Content-Type: application/json" -d "{\"name\":\"concurrent-$i.com\",\"type\":\"domain\"}"
```

```
[8] Done      curl -s -X POST http://localhost:8080/assets -H "Content-Type: application/json" -d "{\"name\":\"concurrent-$i.com\",\"type\":\"domain\"}"
```

```
[9] Done      curl -s -X POST http://localhost:8080/assets -H "Content-Type: application/json" -d "{\"name\":\"concurrent-$i.com\",\"type\":\"domain\"}"
```

```
[10] Done     curl -s -X POST http://localhost:8080/assets -H "Content-Type: application/json" -d "{\"name\":\"concurrent-$i.com\",\"type\":\"domain\"}"
```

```
[11] Done     curl -s -X POST http://localhost:8080/assets -H "Content-Type: application/json" -d "{\"name\":\"concurrent-$i.com\",\"type\":\"domain\"}"
```

```
[12] Done     curl -s -X POST http://localhost:8080/assets -H "Content-Type: application/json" -d "{\"name\":\"concurrent-$i.com\",\"type\":\"domain\"}"
```

```
[13] Done     curl -s -X POST http://localhost:8080/assets -H "Content-Type: application/json" -d "{\"name\":\"concurrent-$i.com\",\"type\":\"domain\"}"
```

```
[14] Done      curl -s -X POST http://localhost:8080/assets -H "Content-Type:
application/json" -d '{"name\":"concurrent-$i.com\","type\":"domain\}"

[15] Done      curl -s -X POST http://localhost:8080/assets -H "Content-Type:
application/json" -d '{"name\":"concurrent-$i.com\","type\":"domain\}"

[16] Done      curl -s -X POST http://localhost:8080/assets -H "Content-Type:
application/json" -d '{"name\":"concurrent-$i.com\","type\":"domain\}"

[17] Done      curl -s -X POST http://localhost:8080/assets -H "Content-Type:
application/json" -d '{"name\":"concurrent-$i.com\","type\":"domain\}"

[18] Done      curl -s -X POST http://localhost:8080/assets -H "Content-Type:
application/json" -d '{"name\":"concurrent-$i.com\","type\":"domain\}"

[19]- Done      curl -s -X POST http://localhost:8080/assets -H "Content-Type:
application/json" -d '{"name\":"concurrent-$i.com\","type\":"domain\}"

[20]+ Done      curl -s -X POST http://localhost:8080/assets -H "Content-Type:
application/json" -d '{"name\":"concurrent-$i.com\","type\":"domain\}"
```

a@DESKTOP-R4B2HM0 MINGW64 ~/Desktop/session2-basic-api

\$ wait

a@DESKTOP-R4B2HM0 MINGW64 ~/Desktop/session2-basic-api

\$ curl http://localhost:8080/assets/count

```
{"count":28,"filters":{}}
```

```
MINGW64:/c/Users/a/Desktop/session2-basic-api
Content-Type: application/json" -d "{\"name\":\"concurrent-$i.com\",\"type\":\"d
omain\"}"
[18] Done curl -s -X POST http://localhost:8080/assets -H "
Content-Type: application/json" -d "{\"name\":\"concurrent-$i.com\",\"type\":\"d
omain\"}"
[19]- Done curl -s -X POST http://localhost:8080/assets -H "
Content-Type: application/json" -d "{\"name\":\"concurrent-$i.com\",\"type\":\"d
omain\"}"
[20]+ Done curl -s -X POST http://localhost:8080/assets -H "
Content-Type: application/json" -d "{\"name\":\"concurrent-$i.com\",\"type\":\"d
omain\"}"

a@DESKTOP-R4B2HM0 MINGW64 ~/Desktop/session2-basic-api
$ wait

a@DESKTOP-R4B2HM0 MINGW64 ~/Desktop/session2-basic-api
$ curl http://localhost:8080/assets/count
{"count":28,"filters":{}}

a@DESKTOP-R4B2HM0 MINGW64 ~/Desktop/session2-basic-api
$ ^C

a@DESKTOP-R4B2HM0 MINGW64 ~/Desktop/session2-basic-api
$ |
```